

Exam Date & Time: 13-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS -JANUARY 2024

SUBJECT: ECE 3154- MICROWAVE ENGINEERING

MICROWAVE ENGINEERING [ECE 3154]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Missing data may be suitably assumed.

- 1A) A 90 W power source is connected to the input of directional coupler with $C=20\text{dB}$, $D=35\text{ dB}$ and an insertion loss of 0.5 dB. Find the output powers at through, coupled, and isolated ports. Assume all the ports are matched (4)
- 1B) Explain the working principle of Hybrid Tee with neat diagram and derive its S matrix (4)
- 1C) A matched isolator has an insertion loss of 0.3 dB and isolation of 30 dB. Find its S matrix (2)
- 2A) Explain the working principle of four port faraday rotation circulator with diagram and list its two applications (4)
- 2B) Explain the bunching process in two cavity klystron amplifier and obtain the expression for bunching parameters (4)
- 2C) With a neat diagram, explain the amplification process in helical TWT (2)
- 3A) The radiation intensity of an Antenna is $\cos^4\theta$ for the $0 \leq \theta \leq 90^\circ$ with $0 \leq \Phi \leq 360^\circ$. Find maximum effective aperture using all the formulae for $f = 10\text{ GHz}$. (4)
- 3B) Show that the maximum directivity of small dipole antenna is 1.5 (4)
- 3C) Discuss any two limitations of vacuum tubes at high frequency with proper justification. (2)
- 4A) Derive the vector wave equation in terms of vector potential (F) and magnetic current density (M) and obtain the solution of it. (4)
- 4B) Show that the directive gain of the Hertzian dipole is

$$G_d(\theta, \varphi) = 1.5 \sin^2 \theta \text{ and that of the half wave dipole is } G_d(\theta, \varphi) = 1.64 \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}. \quad (4)$$

- 4C) A short dipole of length 0.1λ is kept symmetrically about the origin, oriented along the z-direction and radiating 1 kW power into free space. Calculate the power density at $r = 1\text{ km}$ along $\theta = 45^\circ$ and $\varphi = 90^\circ$. (2)
- 5A) For the two-element array system, the two antennas are separated by a distance d . Find the nulls of the total field when $d = \lambda/4$ and the progressive phase shift (4)

- i. $\beta=0$
- ii. $\beta=+\pi/2$
- iii. $\beta=-\pi/2$

- 5B) Design a rectangular microstrip antenna using a substrate (RT/Duroid 5880) with dielectric constant of 2.2, $h = 0.1588$ cm, to resonate at 10 GHz. (4)
- 5C) Describe briefly: mm-wave antenna (2)

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